lifetime of work, all the risks she is prepared to take, and all the energy and effort that she is able to put into the welfare of children. How should a young female, setting out on her adult life, invest her life's resources? What would be a wise investment policy for her to follow? We have already seen from the Lack theory that she should not spread her investment too thinly among too many children. That way she will lose too many genes: she won't have enough grandchildren. On the other hand, she must not devote all her investment to too few children—spoilt brats. She may virtually guarantee herself *some* grandchildren, but rivals who invest in the optimum number of children will end up with more grandchildren. So much for even-handed investment policies. Our present interest is in whether it could ever pay a mother to invest unequally among her children, i.e. in whether she should have favourites.

The answer is that there is no genetic reason for a mother to have favourites. Her relatedness to all her children is the same, $\frac{1}{2}$. Her optimal strategy is to invest *equally* in the largest number of children that she can rear to the age when they have children of their own. But, as we have already seen, some individuals are better life insurance risks than others. An under-sized runt bears just as many of his mother's genes as his more thriving litter mates. But his life expectation is less. Another way to put this is that he *needs* more than his fair share of parental investment, just to end up equal to his brothers. Depending on the circumstances, it may pay a mother to refuse to feed a runt, and allocate all of his share of her parental investment to his brothers and sisters. Indeed it may pay her to feed him to his brothers and sisters, or to eat him herself, and use him to make milk. Mother pigs do sometimes devour their young, but I do not know whether they pick especially on runts.

Runts constitute a particular example. We can make some more general predictions about how a mother's tendency to invest in a child might be affected by his age. If she has a straight choice between saving the life of one child or saving the life of another, and if the one she does not save is bound to die, she should prefer the older one. This is because she stands to lose a higher proportion of her life's parental investment if he dies than if his little brother dies. Perhaps a better way to put this is that if she saves the little brother she will still have to invest some costly resources in him just to get him up to the age of the big brother.